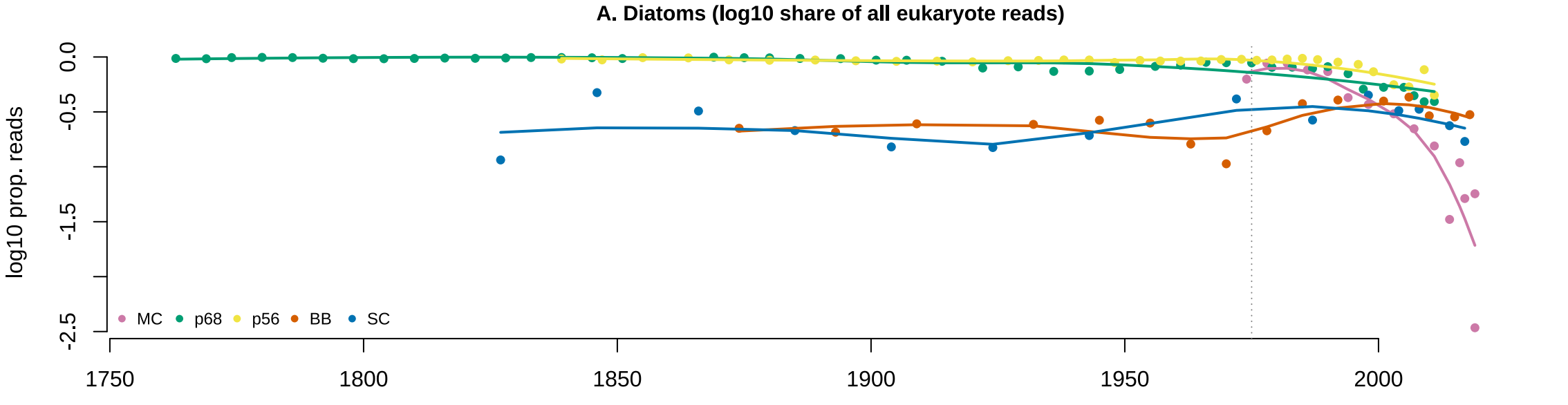
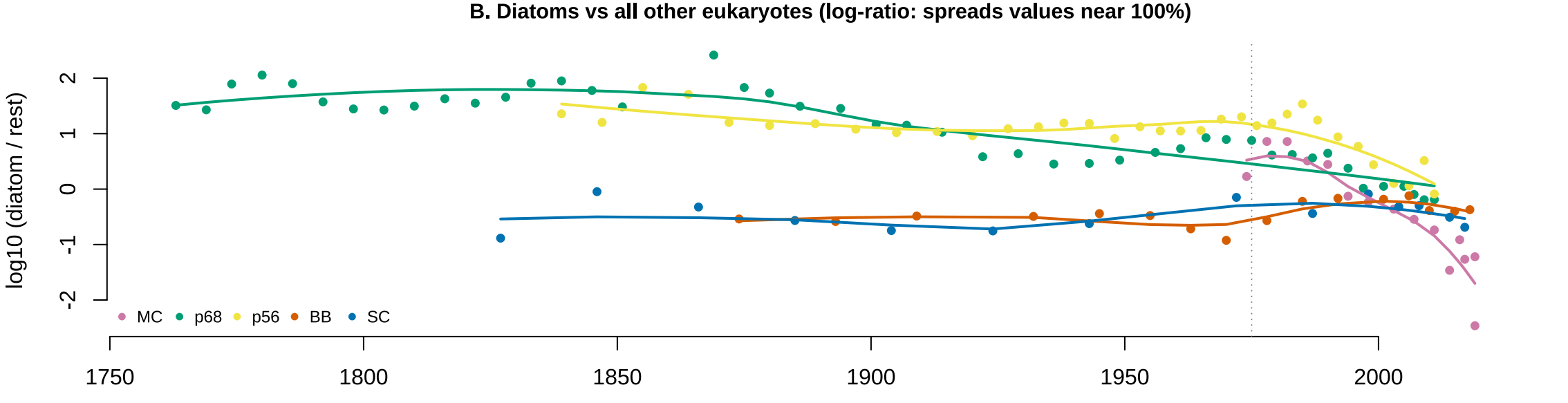
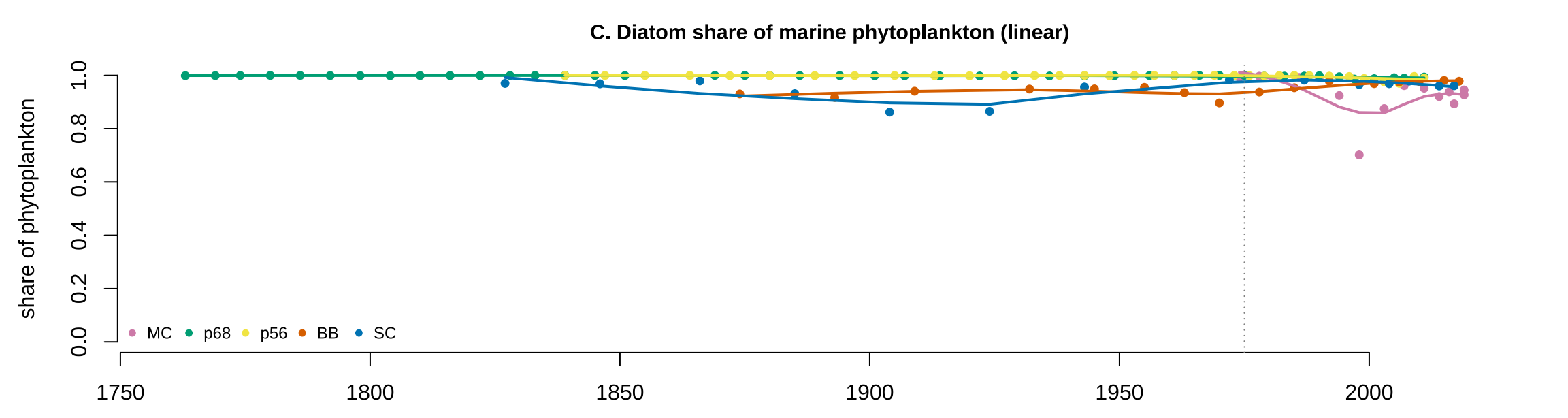
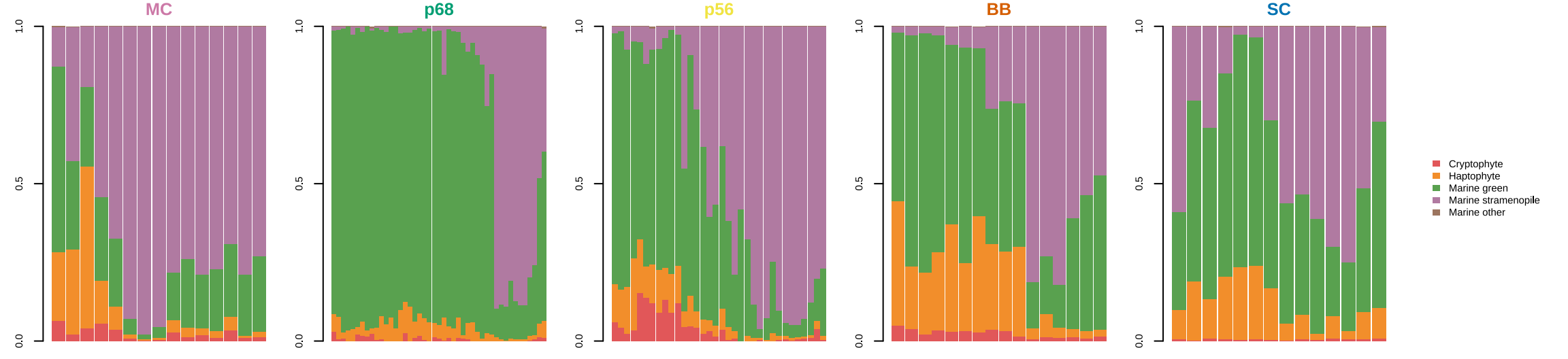
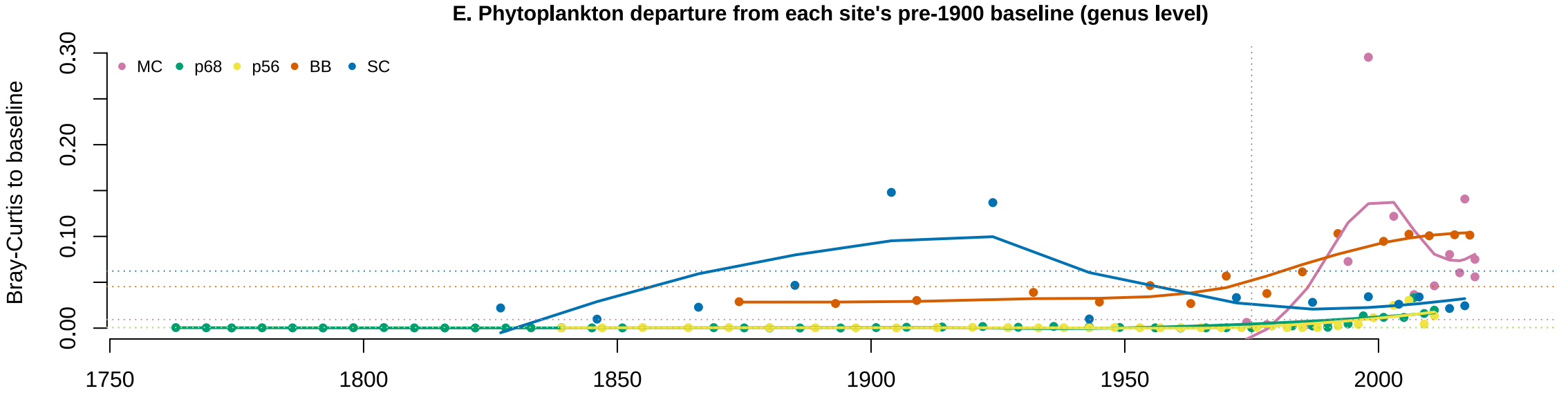

A. Diatoms (log10 share of all eukaryote reads)
log10 prop. reads
0.0
-0.5
-1.5
-2.5
MC
p68
p56
BB
SC
1750
1800
1850
1900
1950
2000

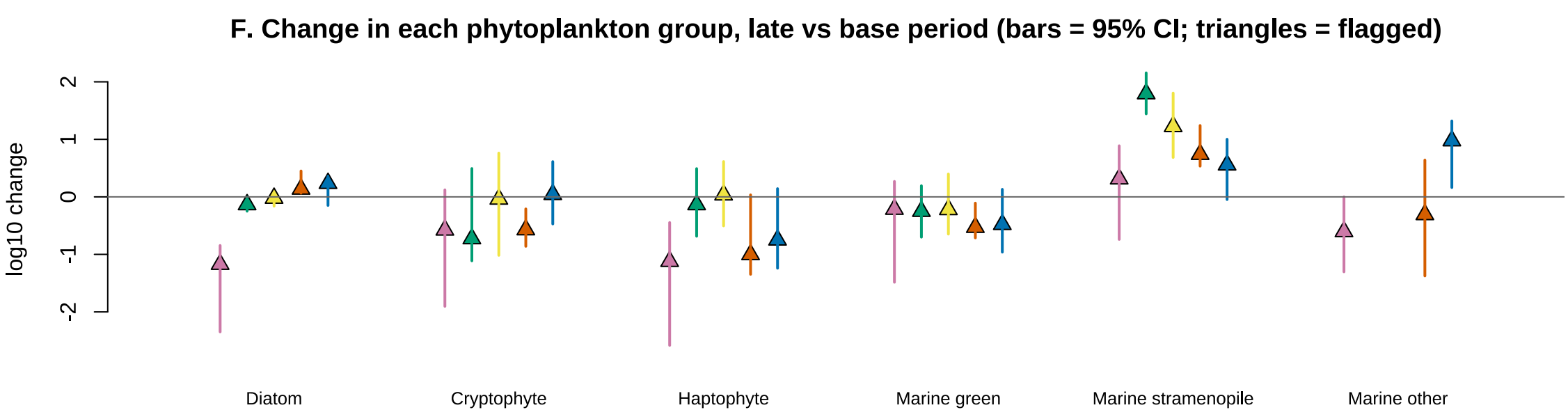

B. Diatoms vs all other eukaryotes (log-ratio: spreads values near 100%)
log10 (diatom / rest)
2
1
0
-1
-2
MC
p68
p56
BB
SC
1750
1800
1850
1900
1950
2000

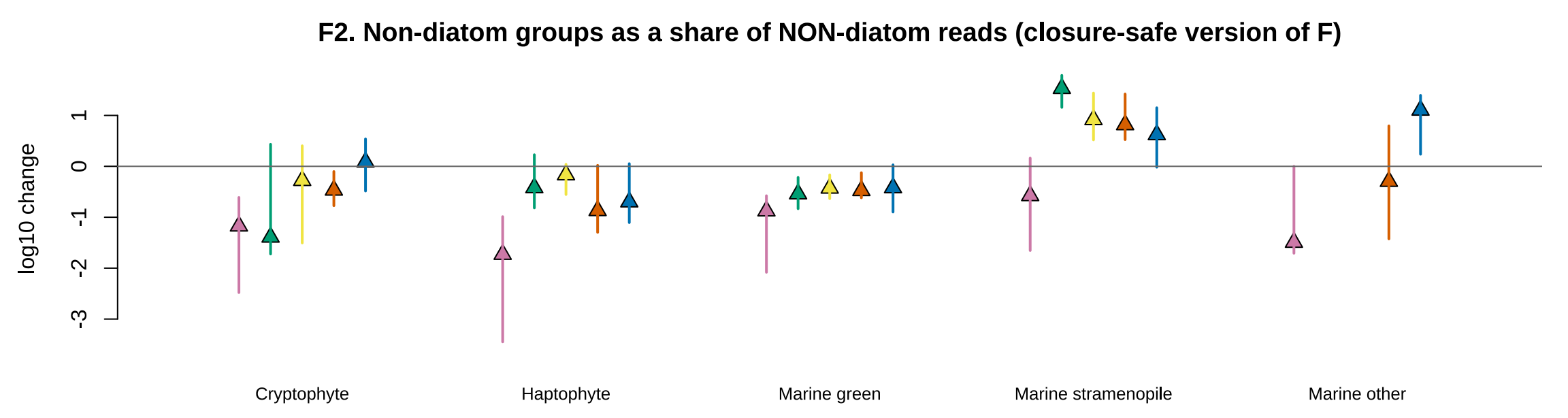

C. Diatom share of marine phytoplankton (linear)
share of phytoplankton
0.0
0.2
0.4
0.6
0.8
1.0
MC
p68
p56
BB
SC
1750
1800
1850
1900
1950
2000

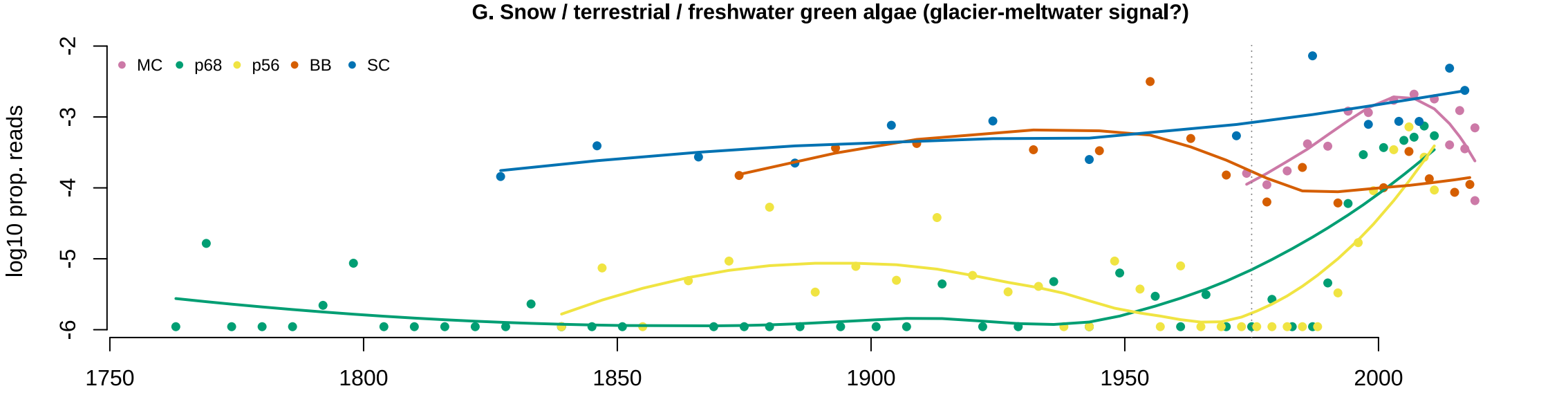
D. Non-diatom phytoplankton composition per site (oldest left)

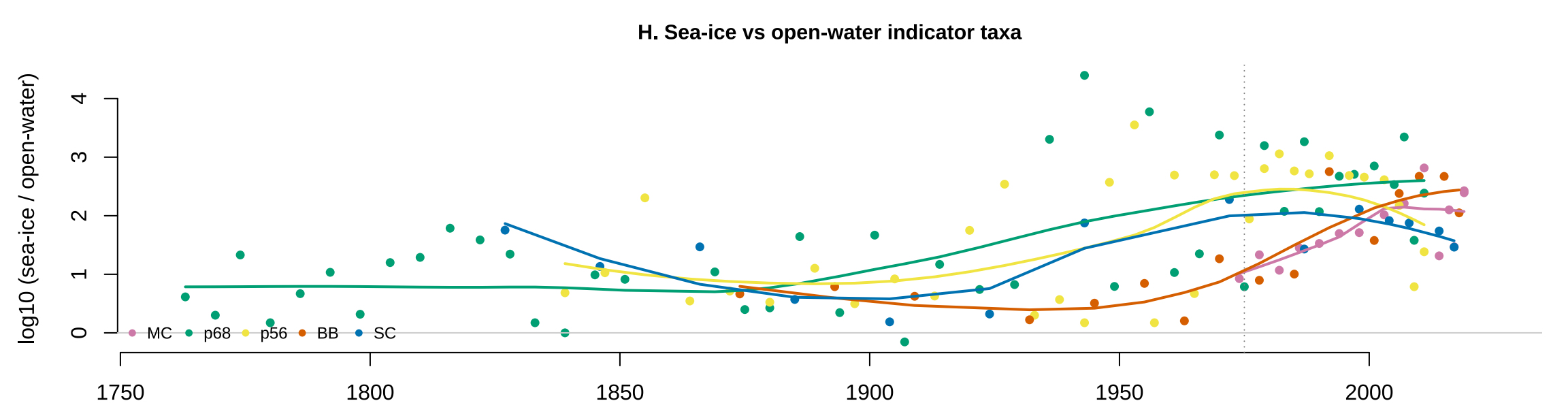


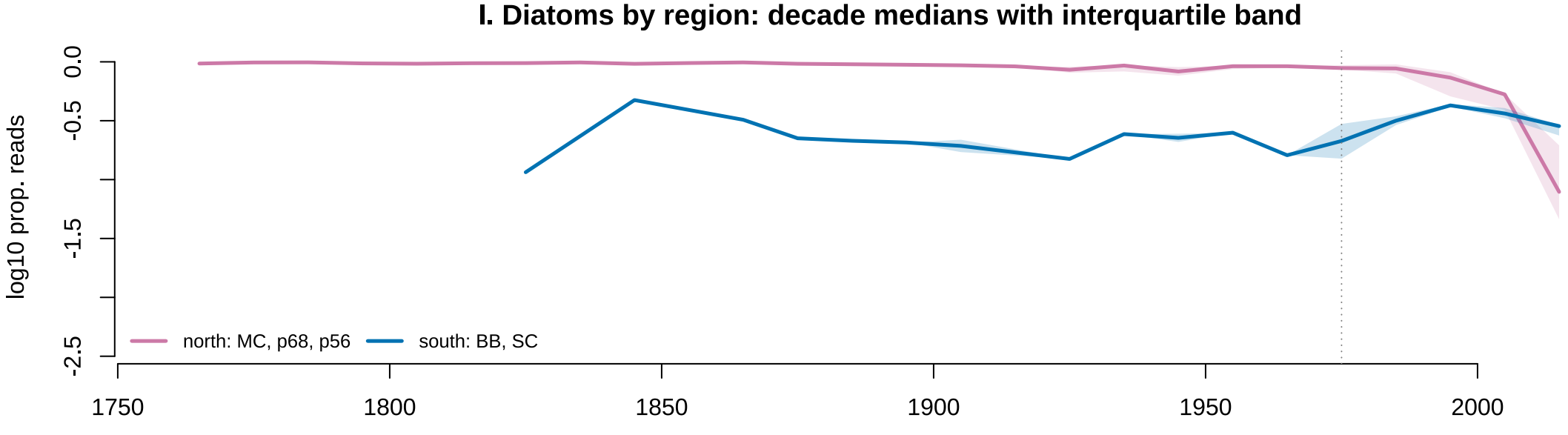

E. Phytoplankton departure from each site's pre-1900 baseline (genus level)
Bray-Curtis to baseline
0.00
0.10
0.20
0.30
MC
p68
p56
BB
SC
1750
1800
1850
1900
1950
2000

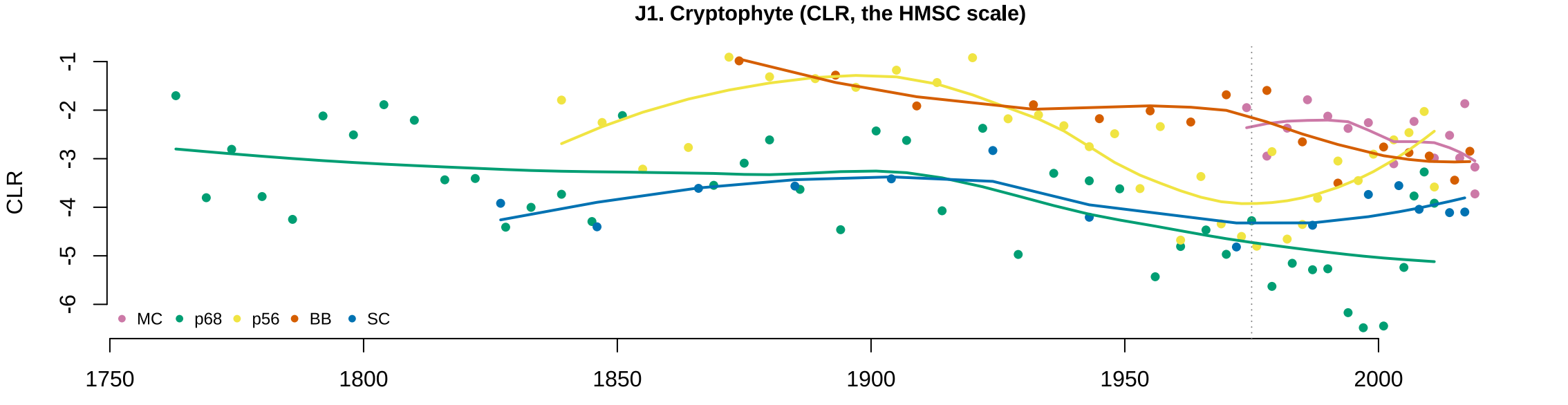

F. Change in each phytoplankton group, late vs base period (bars = 95% CI; triangles = flagged)
log10 change
2
1
0
-1
-2
Diatom
Cryptophyte
Haptophyte
Marine green
Marine stramenopile
Marine other

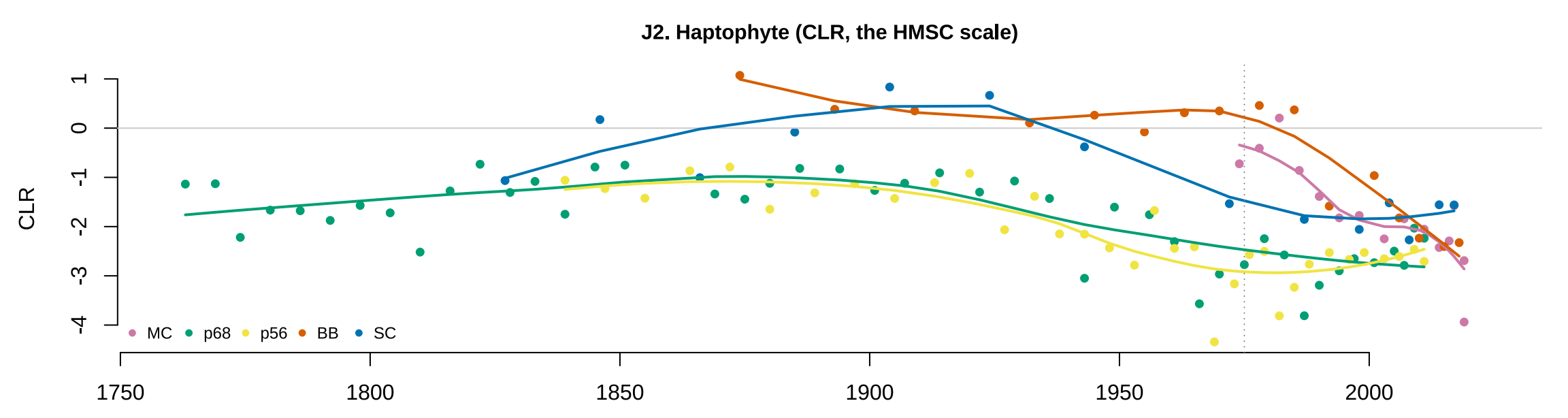

F2. Non-diatom groups as a share of NON-diatom reads (closure-safe version of F)
log10 change
1
0
-1
-2
-3
Cryptophyte
Haptophyte
Marine green
Marine stramenopile
Marine other

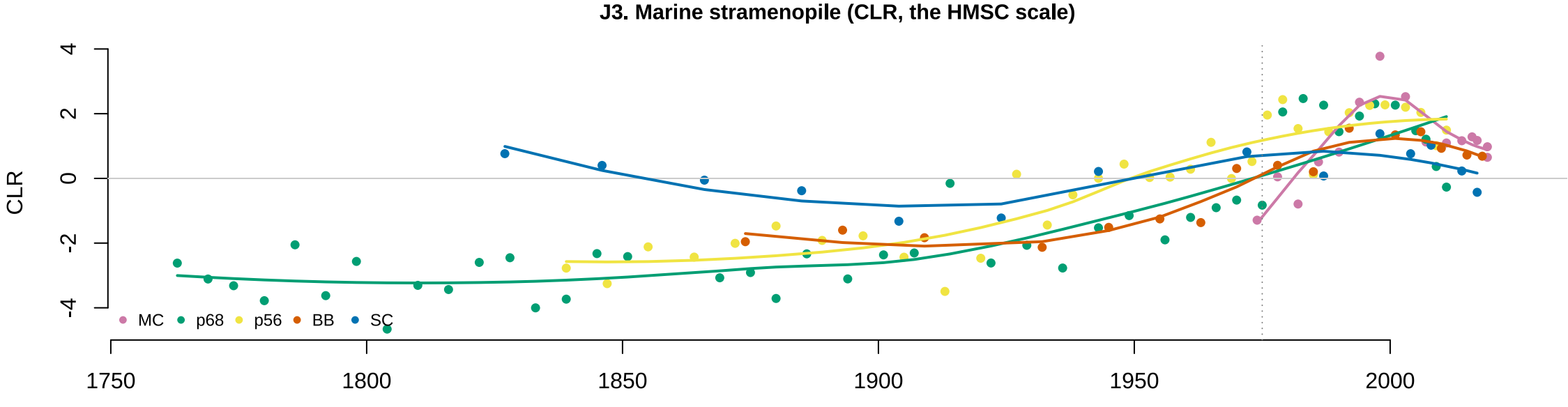

G. Snow / terrestrial / freshwater green algae (glacier-meltwater signal?)
MC
p68
p56
BB
SC
log10 prop. reads
-2
-3
-4
-5
-6
1750
1800
1850
1900
1950
2000

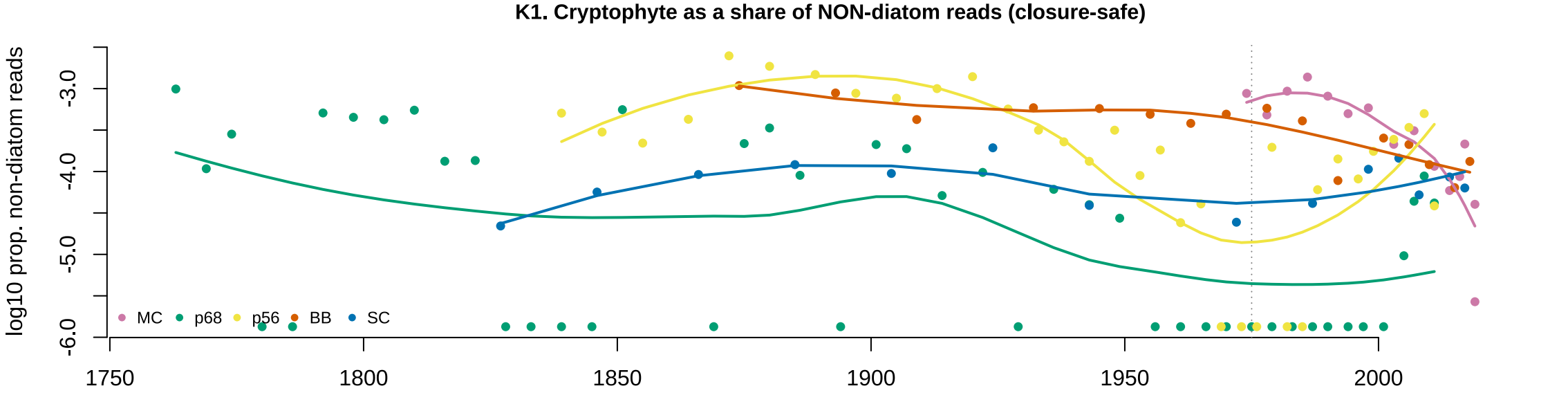

H. Sea-ice vs open-water indicator taxa
log10 (sea-ice / open-water)
0
1
2
3
4
MC
p68
p56
BB
SC
1750
1800
1850
1900
1950
2000

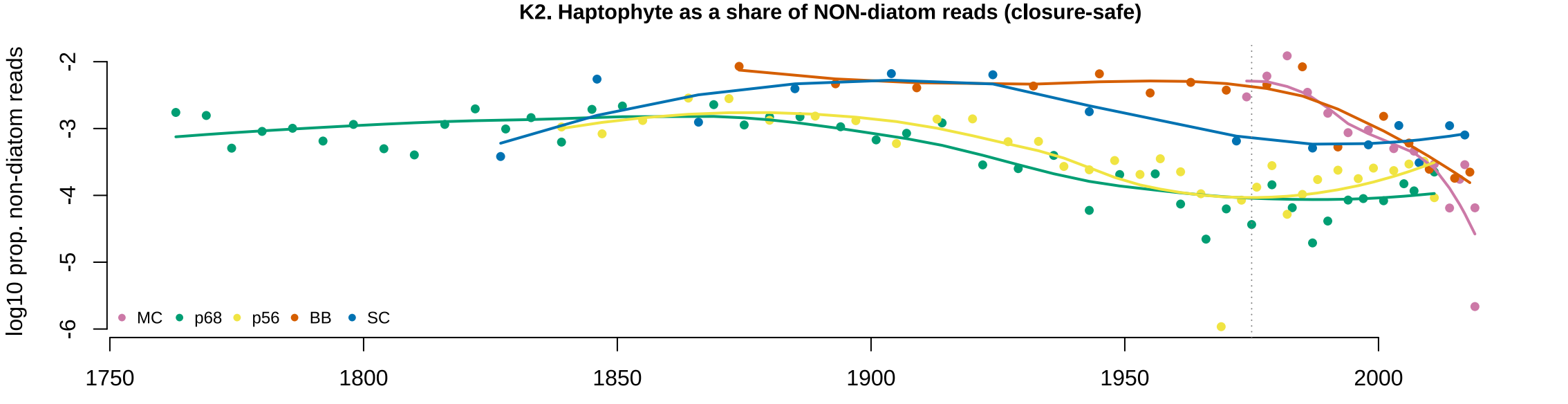

I. Diatoms by region: decade medians with interquartile band
log10 prop. reads
0.0
-0.5
-1.5
-2.5
north: MC, p68, p56
south: BB, SC
1750
1800
1850
1900
1950
2000

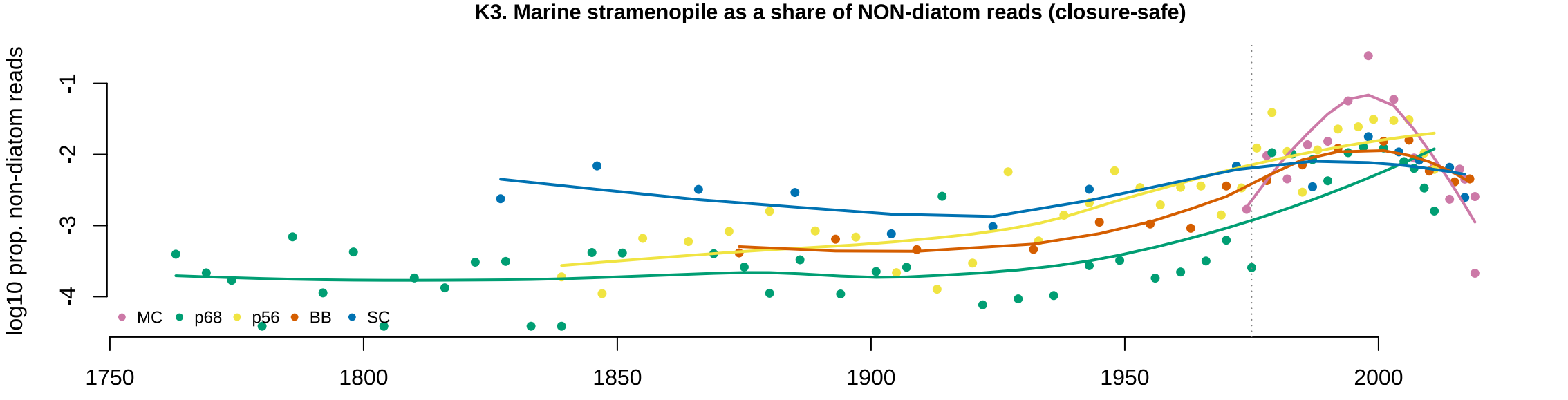

J1. Cryptophyte (CLR, the HMSC scale)
CLR
-1
-2
-3
-4
-5
-6
MC
p68
p56
BB
SC
1750
1800
1850
1900
1950
2000


J2. Haptophyte (CLR, the HMSC scale)
CLR
1
0
-1
-2
-3
-4
MC
p68
p56
BB
SC
1750
1800
1850
1900
1950
2000


J3. Marine stramenopile (CLR, the HMSC scale)
CLR
4
2
0
-2
-4
MC
p68
p56
BB
SC
1750
1800
1850
1900
1950
2000


K1. Cryptophyte as a share of NON-diatom reads (closure-safe)
log10 prop. non-diatom reads
-3.0
-4.0
-5.0
-6.0
MC
p68
p56
BB
SC
1750
1800
1850
1900
1950
2000


K2. Haptophyte as a share of NON-diatom reads (closure-safe)
log10 prop. non-diatom reads
-2
-3
-4
-5
-6
MC
p68
p56
BB
SC
1750
1800
1850
1900
1950
2000


K3. Marine stramenopile as a share of NON-diatom reads (closure-safe)
log10 prop. non-diatom reads
-1
-2
-3
-4
MC
p68
p56
BB
SC
1750
1800
1850
1900
1950
2000